AUTOMATIC IDENTIFICATION OF CLASS COHERENT AND HIGHRARACHIY TO IMPROVE MULTI-CLASS CLASSIFICATION PROBLEM.

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AGENDA

Background of the study

This section will describe highlevel problem statement and motivations for the study

Aims & Objectives

Explains what are the aims and objectives of the study.

Literature Review

Review of techniques availible in news classification and gaps in comparesion with the problem statement

Methodology

This section will explain, proposed solution in detailed to fulfill the gaps identified in LR

Model analysis

Detailed analysis on various subtasks in proposed solution and model performence

Conclusions & Our Contributions

Explain overall finding of the research and elements of our contribution.

BACKGROUND

INTRODUCTION

Information retrieval (IR) is a complex job from text documents.

Document labeling is a common tool for efficient IR on current digital times

Current social media platforms enable users to create #tags for information categorization

CHALLENGES IN NEWS CATEGORIZATIONS

1. Wealth of Information
Increase in news cirulation in the form of digital articles (blogs, micro blogging, news items) as well increase in unorgnized news distribution.

2. Multiple contexts of news
News items may typical representing multiple news context. Identifiying machines to recongnize these classifications became importent

Order of context .3
Identify the context weigths covering in the news item, for example politics and health on same article, weigths will help to determine importent context of the article

Time factor .4
At times, metadata such as classes or #tags need to be dynamically update to match the current situation

BACKGROUND

AIM & OBJECTIVES

The main aim of the study, to propose the method for identify label coherence & hierarchical relationships between the labels. With the intention to compress the dense class labels to the finite set for enabling machine learning model on learning higher quality results without loosing granularity of details present in the news classifications

1. Implement statistical framework for identifying label classifications, through the mixed knowledge of human labeling and unsupervised techniques.

2. To develop a procedure for identifying label hierarchies

3. To implement suitable model development for predicting multi—label and multi-class classifications.

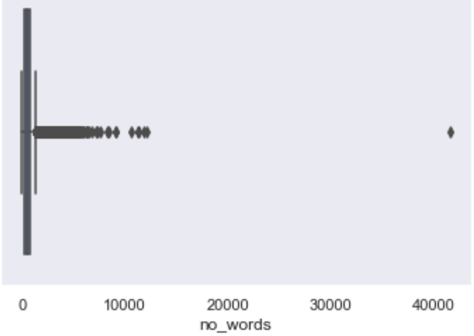
4. To implement neural network for classifying high dense labels as well as finite label sets.

BACKGROUND

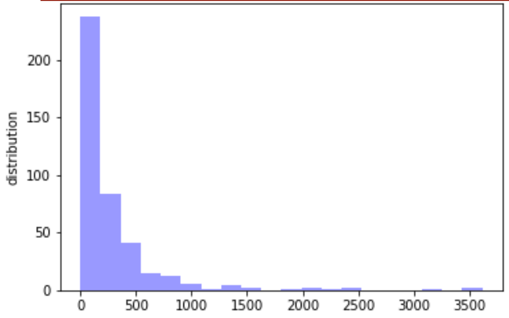
DATA SET DESCRIPTION & EDA

	Metric
Number of News articles	48,968
Number of class labels	441
Average articles word count across a corpus	573
Median article word count	468

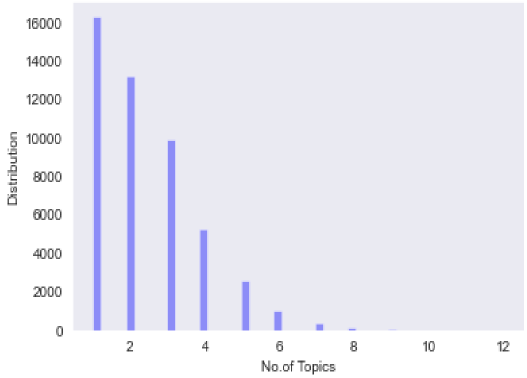
Document Size Distribution



Number of Observations per label



No. of Labels per document



Coherence labels

Label-1	Label-2
investment	investment-&-company-information
investment-&-company-information	money
crime-&-justice	crime-&-punishment
celebrity	hollywood
economics	economy-of-the-united-states
budget,-tax-&-economy	economics
world_news	disasters-&-accidents
company-legal-&-law-matters	laws-and-regulations
military-&-defense	u.s.-military
movies	hollywood

LITERATURE REVIEW

TEXT PROCESSING & FEATURE ENGINEERING

1

Text processing

Natural language text contains several syntatatical & symantical sentence frameing text. That may be importent for language construction. But it create a unnessary complex in document classifications.

Dudhabaware and Madankar, 2015; Lourdusamy and Abraham, 2018, suggested following basic text processing steps will make document classification task easy and help in removing the words that really not helping in constructing news classifications.

- Tokenization
- Removal of Stop words
- Lemmatization and stemming

2

Feature extraction

Feature extraction is one of the critical processing step in NLP tasks. Since it is a complex job for mathamatical models to learn the text information.

R et al., 2018, suggested legacy of few word embedding technique that can be used for creating features from the text documents.

- Bag of Word method
- Term frequency – Inverse frequency (TFIDF)
- Word ending created with the help of other big corpus on n-grams & skip graps.
- Word2vec *Tomas 2013*
- Gove *Jeffrey, Richard , Christopher ,2014*
- Pre-trained vs fine tuned vectorization

3

Topic Modelling

It is way of learning the number of topics present in the document set based on the frequency of words that appeared in mixed set of documents. This will enable to identify the similarities amount of document sets, also enable to extract information asosiated to the perticlural topics through unsupervised apparoch.

- Latent Dirichlet allocation (LDA) *David, Andrew, Michael 2003*
- Top2Vec Angelov 2020

LITERATURE REVIEW

SOLVING MULTI-LABEL LEARNINGS

1

Simple problem transformation technique

Most of the time classification usually refer as binary class classification, however problems like news categories to have document to be represented multiple different type of classes and some times obsevation belongs to the two set of class labels.

(Prajapati et al., 2012) sugested varioation of MLL solutions

1. Simple problem transferomation

   Label powerset, Binary relevance

2. Algorithem tuning approch

   Boostexter, custimization of adaboost Schapire and Singer, 2000

   neural network implementation (BP-MLL) Zhang and Zhou

   SVM kernal optmizers, ML-KNN optmization of nearest neighbor

3. Different activation functions to compute ranking loss between labels, sigmoid over softmax

2

Hierarchical classification

In situations where high dense labels or unorgnized labels present in large extent, heirarachical structure will enable researcher to focus on the details in which suites business requirements.

(Prajapati et al., 2012) sugested varioation of MLL solutions

- Constructions of hierarchical structure

  (Li et al., 2003) uses LDA and topic projection to identify label

similarities. And then based on Euclidean distance Hierarchical Agglomerative Clustering(HAC)

(Soni et al., 2017), developed label coherence matrix based on the frequency of labels together constructed tree structure

- Solving classifications

  Local Classification per Node

  Local Classifier per Parent Node (LCPN)

  Local Classification per Level (LCL)

LITERATURE REVIEW

HIERARCHICAL CLASSIFICATIONS

1. Challenges in Constructing label hierarchies
2. LDA -> Hierarchical clustering by (Li et al., 2003)
3. Label coherence matrix based on the frequency of labels together constructed tree structure by Soni et al., 2017),

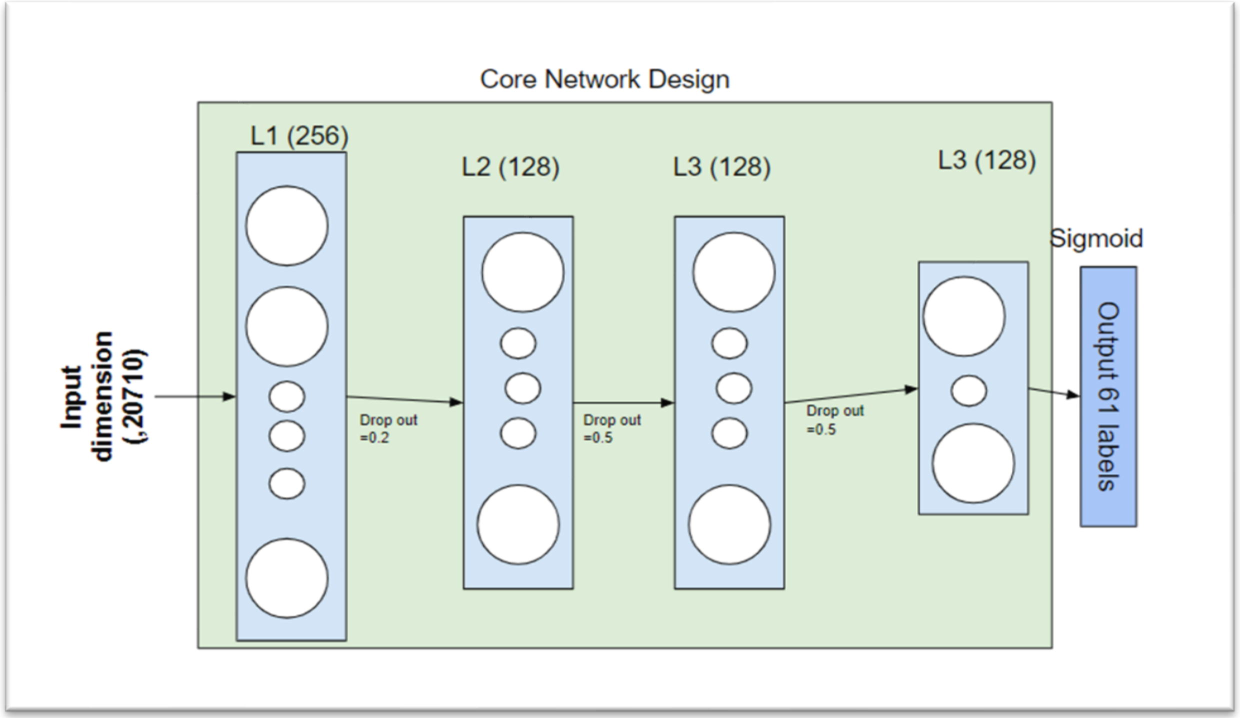
LITERATURE REVIEW

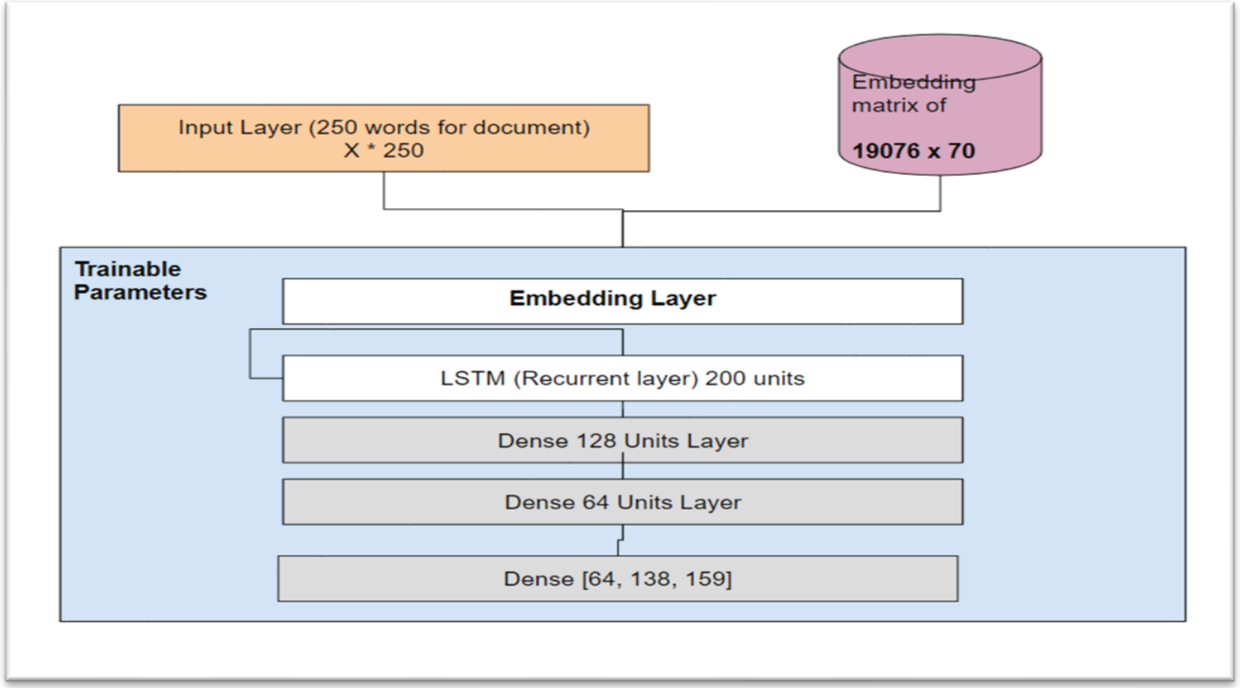
NEURAL NETWORK IMPLEMENTATION FOR MLL

1. Challenges in Text classification through NN
   High dimensions and variable dimensionalities (no.of words)

2. Benefits of RNN kind of LSTM (Minaee et al., 2020)

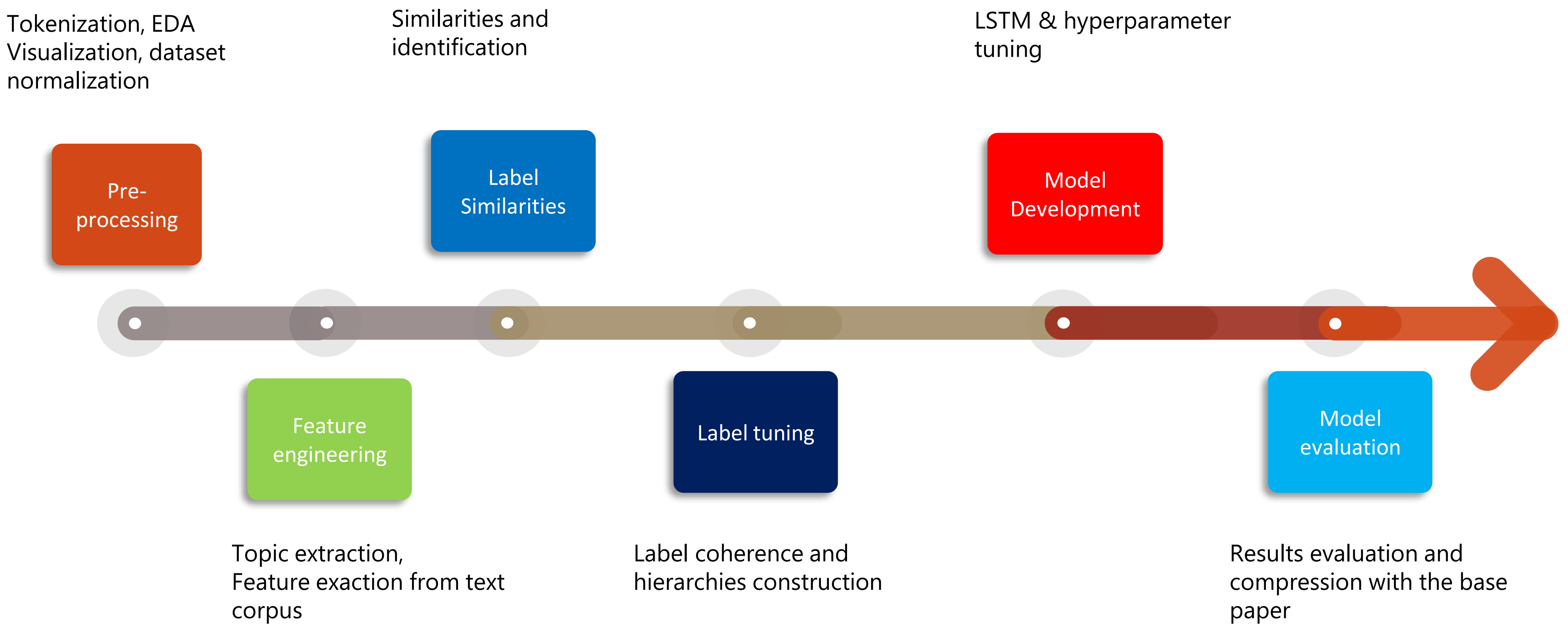
3. Multi Label output through postprocessing





METHODOLOGY

OVERVIEW



METHODOLOGY

METHOD (LABEL TUNING)

Label-1	Label-2
investment	investment-&-company-information
investment-&-company-information	money
crime-&-justice	crime-&-punishment
celebrity	hollywood
economics	economy-of-the-united-states
budget,-tax-&-economy	economics
world_news	disasters-&-accidents
company-legal-&-law-matters	laws-and-regulations
military-&-defense	u.s.-military
movies	hollywood

Label Similarities

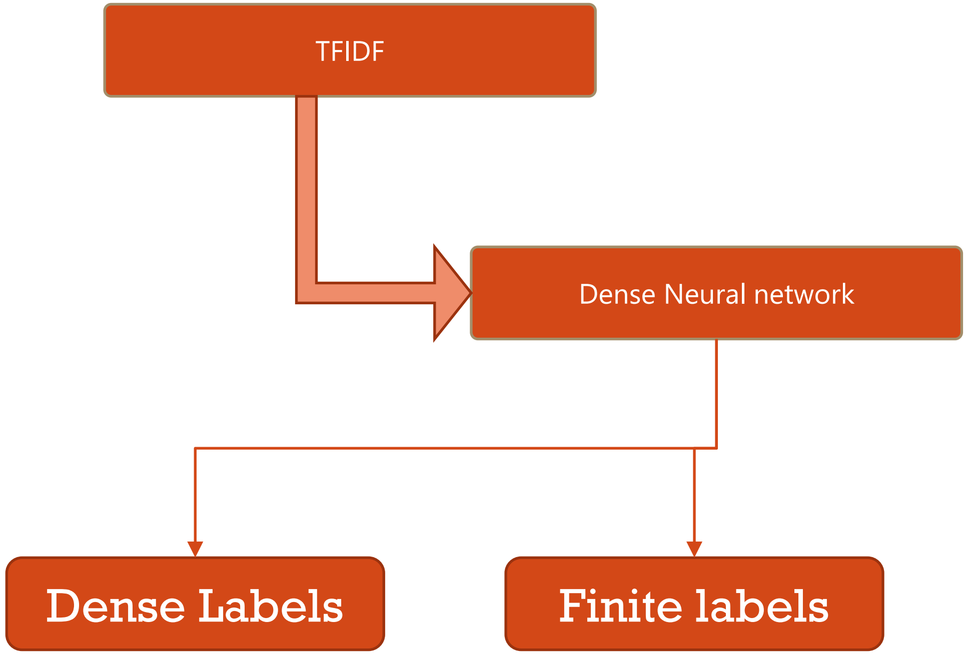
Label Coherence

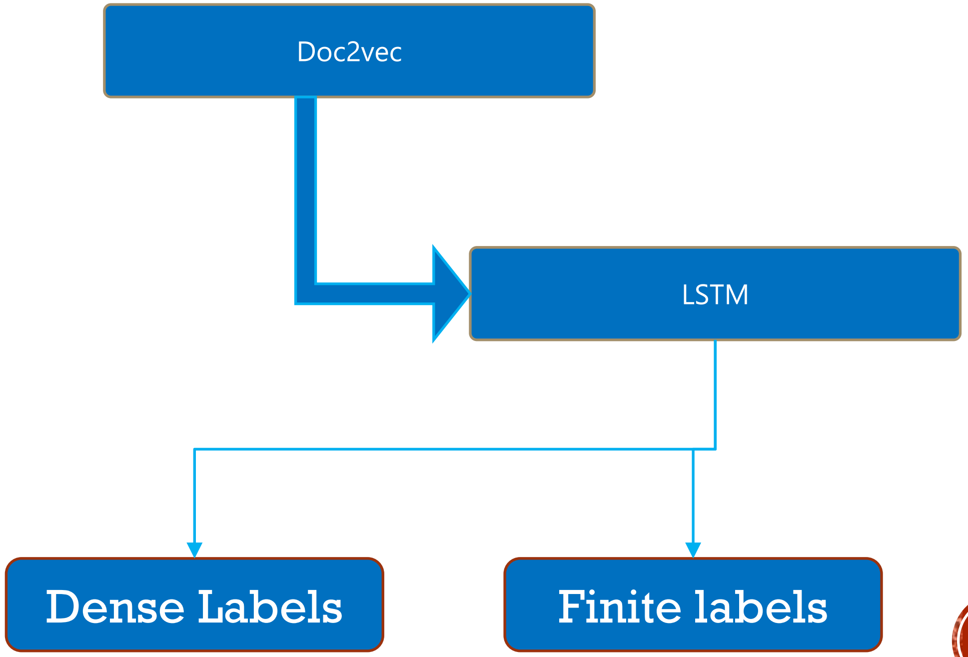
Label hiararchies

METHODOLOGY

PROPOSED METHOD –CONT.. (MODEL DESIGN)

Method -1	Method -2





MODEL ANALYSIS

PRE-PROCESSING

Word Treatment

WORD	Corrected Word(word segmentation)
homeusabusiness	home USA business
writertime	writer time
researchandmarkets	research and markets
prevnextlast	prev next last
firsthighest	first highest

Label Treatment

	Number of Observations
Labels with lesser than 10 observations	35
Labels with lesser than 25 observations	75
Labels with lesser than 50 observations	105
Labels with lesser than 100 observations	167
Labels with lesser than 200 observations	261
Labels with lesser than 500 observations	326
Labels with lesser than 800 observations	388
Labels with lesser than 1000 observations	397

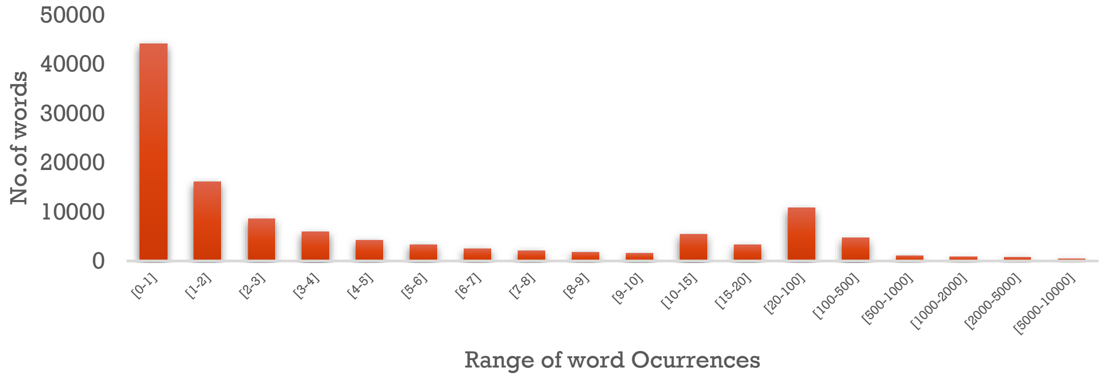
Note: Label treatment after removing labels lesser than 100 observations

Corpus treatment

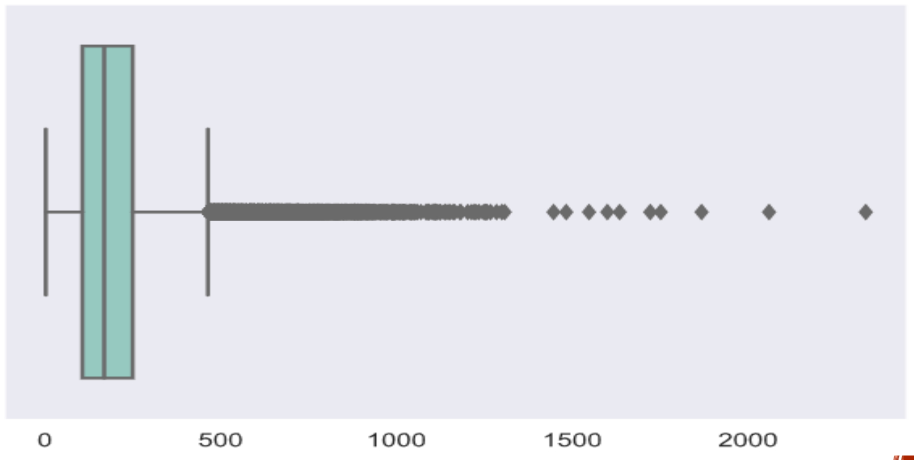
Tokenization

Stop words

Stemming



Document size distribution



MODEL ANALYSIS

FEATURE ENGINEERING

Topics identified throoigh **Top2vec**

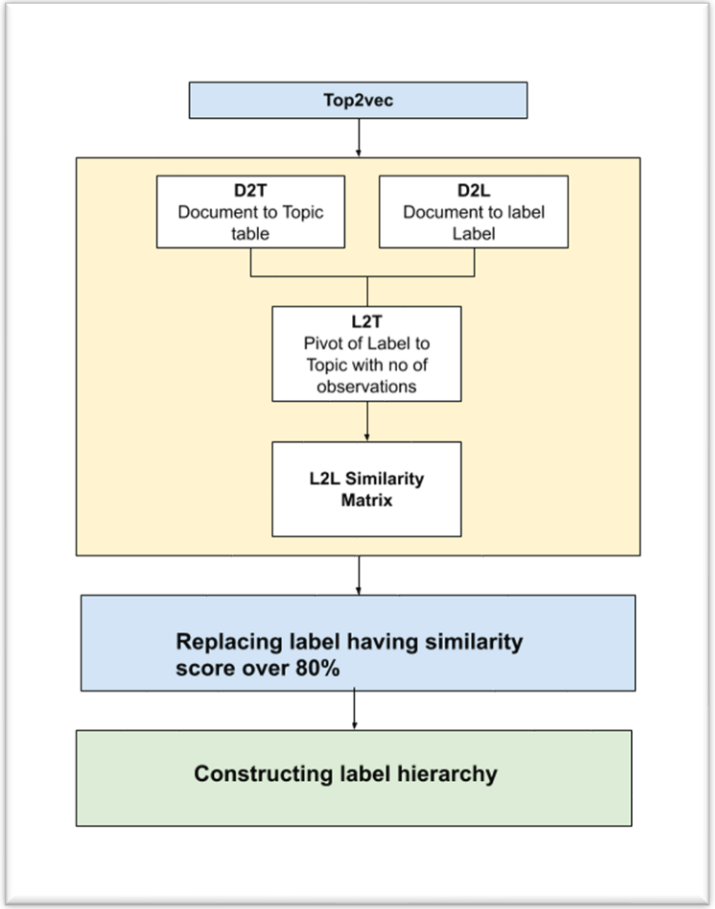
Topic	Top 10 for each topic [Word (word score)]
2	'song(0.82), album(0.8), songs(0.79), albums(0.77), band(0.71), songwriter(0.7), singer(0.66), guitarist(0.65), billboard(0.64), drummer(0.63)
0	prosecutors(0.66), felony(0.64), misdemeanor(0.62), pleaded(0.62), burglary(0.61), sentencing(0.61), aggravated(0.61), charged(0.6), victim(0.6), prosecutor(0.58)
5	democrats(0.8), republicans(0.79), bipartisan(0.77), congressional(0.75), senate(0.74), overhaul(0.72), legislation(0.72), steny(0.71), congress(0.71), pelosi(0.71)
100	pastors(0.77), pastor(0.77), church(0.76), churches(0.75), christ(0.73), congregation(0.69), god(0.68), bible(0.67), baptist(0.64), gospel(0.64)
20	qaeda(0.8), militants(0.78), terrorist(0.74), militant(0.72), attacks(0.71), islamist(0.71), pakistani(0.69), terrorists(0.69), qaida(0.69), al(0.67)

Hyperparameters

Hyperparameter	Value	Description
min_count	50	Ignore all the words that have a frequency lesser than 50
embedding_model	Word2vec	Model internally train word2vec from scratch
speed	Learn	This option will enable to learn good quality of words with fewer computation requirements.

MODEL ANALYSIS

LABEL SIMILARITIES



topic_id	0	1	2	3	4	5	6	7	8	9	...	362	363	364	365	366	367	368	369	370	371
ylabel_modified																					
investment	2.0	0.0	0.0	1.0	139.0	1.0	2.0	2.0	31.0	165.0	...	5.0	1.0	5.0	1.0	12.0	0.0	0.0	0.0	2.0	7.0
investment-&-company-information	3.0	0.0	0.0	0.0	156.0	1.0	2.0	2.0	34.0	174.0	...	5.0	1.0	5.0	1.0	13.0	0.0	0.0	0.0	3.0	7.0

2 rows × 372 columns

	level_1	level_2	similarity_score
50636	space-&-astronomy	space-exploration	0.999927
50881	space-exploration	space-&-astronomy	0.999927
34085	mma	mixed-martial-arts	0.999871
33840	mixed-martial-arts	mma	0.999871
12102	crime-&-punishment	crime-&-justice	0.999735
...	...	...	...
2956	audio-technology	adoption	0.000000
29508	juvenile-delinquency	vacation-packages	0.000000
4319	bathroom	mixed-martial-arts	0.000000
29509	juvenile-delinquency	video-games	0.000000
33999	mma	cultural-groups	0.000000

MODEL ANALYSIS

PSEUDOCODE FOR LABEL HIERARCHIES & COHERENCES

```
Var higher_similarties = from L2L where similarity_score >= 0.8

Labels_to_be_replaced = { }

#identify lbels that need to be replaced
For row in higher_similarties:

        No_of_obsevations_label1 = count ( from D2L where label == row.level1 )
        No_of_obsevations_label2 = count ( from D2L where label == row.level2 )

        If No_of_obsevations_label1 > No_of_obsevations_label2:

                Labels_to_be_replaced [ level2 ] = Labels_to_be_replaced [ level1 ]
        Else:

                Labels_to_be_replaced [ level1 ] = Labels_to_be_replaced [ level2 ]
        Endif

End for

#execute replacement
For item in Labels_to_be_replaced:

        Y_train.replace(item) = Labels_to_be_replaced[item]
        Y_test.replace(item) = Labels_to_be_replaced[item]
End for

#replace duplicates if any because example same article was tagged ealiear by movies and oscar, and our process replaced Oscar with movies, we have now attached movies twice
Y_train = set(y_train)
Y_test = set(y_test)
```

```
# Prepare the similarity scores for the remaining labels in the form of Label1, Label2, similarity score, no.of.obsevations in level 1, number of obsevartions in level 2]

L2L = [Leve1, Leve2, Simiarity_score, No_of_level1_Obsevrations, No_of_leve2_obsevations]

L2l.hirarchy_rank =-1
Rank=0

Count_of_unidentified_ranks = count( where L2l. hirarchy_rank =-1 )
_levels =[root]

While ( Count_of_unidentified_ranks)

        If L2L[parent == _levels ]. hirarchy_rank = rank

        _levels = identified all higher order levels from L2L

        Rank+=1

End while
```

Movies 500	Movie review 200	98%

MODEL ANALYSIS

LABEL COHERENCE AND HIERARCHIES

Redundant labels

Coherent label	Replaced with
movie-reviews	2016_oscars
2016_oscars	movies
pregnancy-termination-services	abortion_debate
NFL	American-football
animal-science	living-nature
armed-forces	military-&-defense
auto-insurance	politics
nail-art	beauty
nail-polish	beauty
skin-care	beauty
bedding-&-linens	furniture
security-upgrades-&-downgrades	bonds
budget,-tax-&-economy	economics
celebrity	Hollywood
Christian	Christianity
glbt-issues	civil-rights
marriage	civil-rights
clothes-&-apparel	fashion
painting	collecting
collecting	visual-arts

Created Label hierarchies

Level-1	Level-2	Level-3
61	77	23

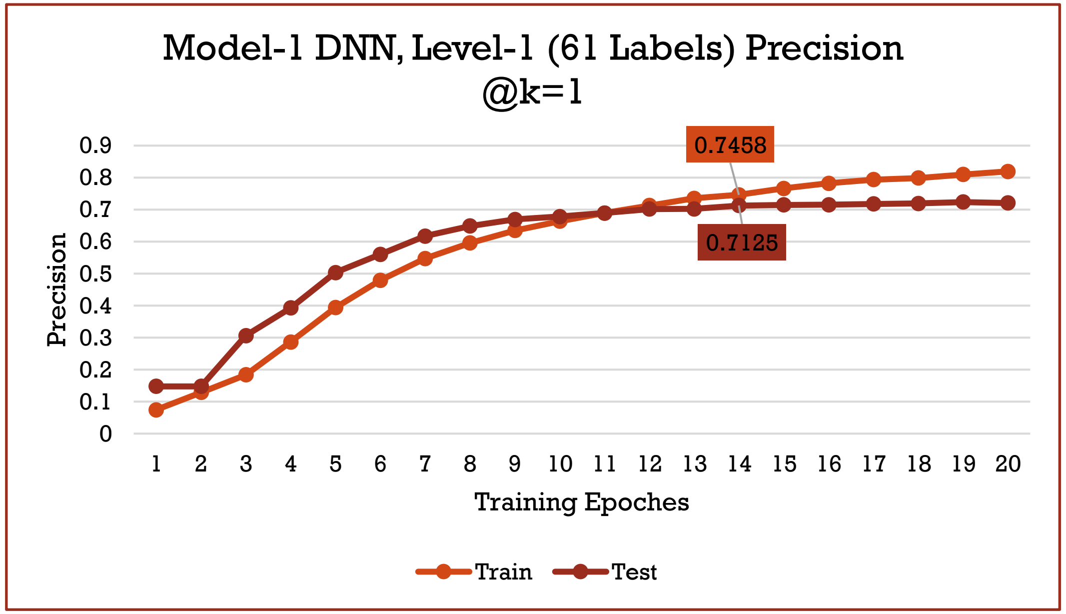
Parent child labels

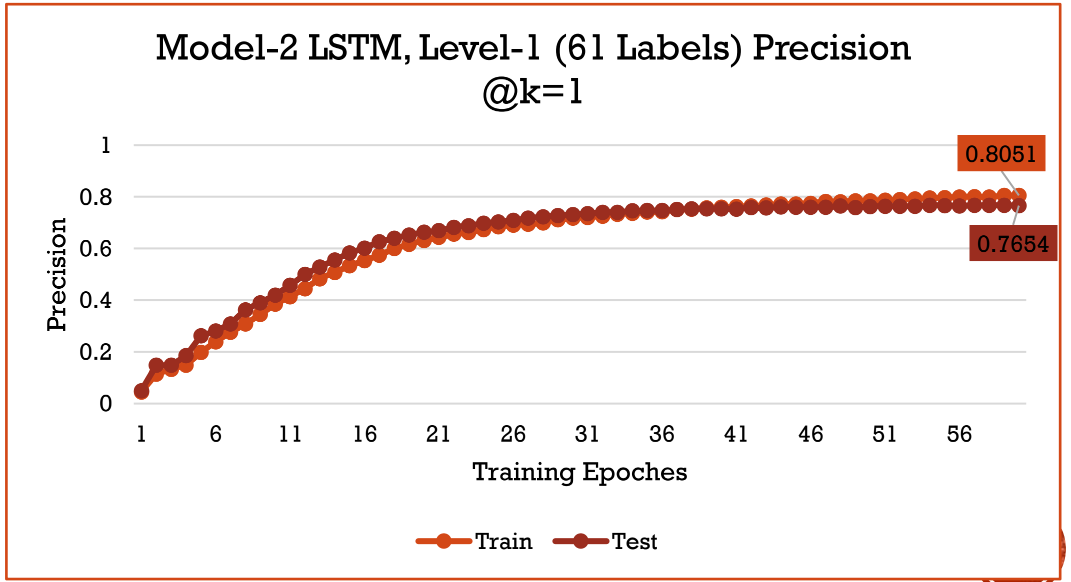
Example	Hierarchy (from bottom to top)
1	nuclear-policy -> foreign-policy -> unrest,-conflicts-&-war -> root
2	Bonds -> stock_market -> investment -> root
3	Toys -> parenting -> root
4	financial-fraud-prevention -> internet-&-networking-technology -> business-services-&-activities -> root
5	Racism -> ethnic-group -> root
6	unemployment-issues -> economics ->root
7	organized-crime-> -> crime-&-justice -> root
8	Addiction -> crime-&-justice -> root

MODEL ANALYSIS

MODEL RESULTS

Remarks	Number of Labels
Total Number of distant labels in the initial database provided by human labeling.	413
Labels identified has redundant (withy higher similarity score)	66
The final set of labels framed to give a high level of granular details on the topics (Level-1)	61
Set of labels having a detailed level of granular details on each topic (Level-1 & Level -2 together)	138
Set of labels having a more detailed level of granular details on each topic (Level -1, Level -2 and Level-3 together)	160





MODEL ANALYSIS

MODEL RESULTS COMPARISONS

MODEL	Metric	Number of Labels	Test Data set Score
BASE PAPER	Precession @k =1	All Tags	0.6702
DNN	Precession @k =1	61 (Level-1)	0.7277
DNN	Precession @k =2	61 (Level-1)	0.4954
DNN	Precession @k =3	61 (Level-1)	0.3712
DNN	Precession @k =1	138 (Level 1 + Level 2)	0.5727
DNN	Precession @k =2	138 (Level 1 + Level 2)	0.4178
DNN	Precession @k =3	138 (Level 1 + Level 2)	0.3278
DNN	Precession @k =1	159 (Level 1 + Level 2 + Level 3)	0.5761
DNN	Precession @k =2	159 ((Level 1 + Level 2 + Level 3)	0.4248
DNN	Precession @k =3	159 ((Level 1 + Level 2 + Level 3)	0.3342
LSTM	Precession @k =1	61 (Level-1)	0.7654
LSTM	Precession @k =2	61 (Level-1)	0.5329
LSTM	Precession @k =3	61 (Level-1)	0.4022
LSTM	Precession @k =1	138 (Level 1 + Level 2)	0.6732
LSTM	Precession @k =2	138 (Level 1 + Level 2)	0.4955
LSTM	Precession @k =3	138 (Level 1 + Level 2)	0.3880
LSTM	Precession @k =1	159 (Level 1 + Level 2 + Level 3)	0.6167
LSTM	Precession @k =2	159 ((Level 1 + Level 2 + Level 3)	0.4519
LSTM	Precession @k =3	159 ((Level 1 + Level 2 + Level 3)	0.3535

CONCLUSIONS & OUR CONTRIBUTIONS

RESEARCH FINDINGS

01

LSTM, trained through word2vec outperformed DNN and P@k =1 is giving better results.

02

The LSTM model for a finite set of labels (61 labels) gives the results of P@k=1 is 0.76, which is approximately 13% higher score than the previous researcher's benchmark scores.

03

Automatically identified labels that can target the business research separately for finite set of labels of 61 and dense set of labels for 156

CONCLUSIONS & OUR CONTRIBUTIONS

OUR CONTRIBUTION & FUTURE RECOMMENDATIONS

1. Mixed models of news agencies information with social media tags could potential help to generate deeper hierarchies

2. Single model for to predict all levels at the same time

THANK YOU

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